

ASEN 6519 AI for Aerospace Systems

Final Project

Assigned: Thursday, November 20, 2025

Due: 11:59 PM, Thursday, December 11, 2025

The goal of this **group assignment** is to implement an end-to-end artificial intelligence architecture for controlling a team of aerial robots. The assignment is based on the two main topics of *swarming control* and *large language models* identified by the class in the prior portion of the course. The assignment consists of a written report in the form of a conference paper and video(s) demonstrating the final system integration. A single assignment will be submitted for the entire class. It should be delivered via email to eric.frew@colorado.edu by 11:59PM on Thursday, December 11, 2025. Note: if the files are too large to send via email, a link to a document in cloud storage is sufficient.

1 Paper

The final report should be written as a conference paper. There is no explicit page limit requirement, but with the format given below I expect the paper to be 15-40 pages in length. Suggestions are also given for the length of each section. The paper must include the following sections:

- **Introduction:** Introduce the problem being addressed, its motivation, and related work. This section should enable the reader to understand the context of your paper. Discussion of related work is of primary importance. The approach of the paper should be introduced at a high level, leaving details of the individual components to the remaining sections. A roadmap to the paper may be introduced at the end of this section. This section should be 1-2 pages in length.
- **Overview:** Provide an overview of the AI architecture that will be implemented. Describe the main components, the interface between them, and major implementation design challenges. This section should have at least one block diagram of the overall system. This section should be 1-3 pages.
- **Large Language Model:** Describe the details of the large language model and its implementation. Include a description of the main model and software tools used. Include a discussion of the strategy for prompt engineering and scenario generation. Include results that showcase the behavior and performance of the LLM that was designed. These results should be different than the end-to-end results in the Experimental Results section, i.e. that could include more complex examples. Use the Appendix to provide additional data that is important but would disrupt the flow of the section.
- **Swarm Control:** Describe the details of the swarm control laws and their implementation. Include a description of all major assumptions, mathematical formulation of the control algorithms, and analysis of any guaranteed properties (use references heavily for this part). Provide results that showcase the behavior and performance of the swarm control laws. These results should be different than the end-to-end results in the Experimental Results section, i.e. that could include more complex examples. Use the Appendix to provide additional data that is important but would disrupt the flow of the section.
- **Robotic Platform:** Describe the simulation environment and platform models used for the final end-to-end experiments. Describe the software architecture for the different components. Also describe the details of the main elements, e.g. the dynamics of the uncrewed aircraft.

- **Experimental Results:** This section will include the main results of the paper. Describe the outcomes of the experiment(s). Include figures and tables that showcase the main results. An important rule of thumb when writing a paper: if a figure, image, or table of results deserves to be in a paper, it deserves several sentences describing the result. Be explicit and direct stating what a reader should notice about a figure.
- **Conclusion:** This section should restate the main goals of the paper and summarize the results. Basic conclusions drawn from the paper should be reiterated here. Brief remarks on future work can be included in the conclusion.
- **Acknowledgements:** Acknowledge the contribution of each student to the assignment by listing each student's role (see the Division of Effort section).
- **References:** A critical component of research publications is the proper citation of other works used in developing the ideas presented in the manuscript. All citations must be properly referenced and formatted.
- **Appendix:** Include information that is important to understand the entire system but would disrupt the flow of the paper, e.g. the all the prompts for the LLM or settings for the robotic platforms.

1.1 Paper Format

The submitted paper must conform to the AIAA Technical Journals format. Instructions and templates can be found online, including here:

<https://www.overleaf.com/latex/templates/preparation-of-papers-for-aiaa-technical-journals/mqgbqqvyhtm>.

I strongly suggest you use the AIAA template files from the beginning as it will save you significant time formatting your manuscript.

1.2 Document Preparation

I also strongly recommend that students use LaTeX to write this paper. There are many good resources online that can get you started. Overleaf.com (<https://www.overleaf.com>) is an excellent online tool for writing LaTeX documents collaboratively.

2 Video

In addition to the paper, the class should submit one or more videos that shows the end-to-end performance of the system. This video should highlight the behavior of each component, showing how it contributes to the final behavior. The video should culminate with at least one example of the swarm accomplishing a non-trivial mission based on a prompt to the LLM. The video DOES NOT need to be polished. Simple text title slides between sections of the video is sufficient, though voice over or text descriptions throughout are welcomed.

3 Division of Effort

All students are expected to make roughly equal contributions to the final report. To insure some level of shared responsibility, the assignment is divided into multiple roles. **Each student is expected to fulfil one and only one of these roles.** Students are able to help one another, but the student assigned to a role has final responsibility for that part of the report.

1. Introduction and System Overview: Provide text and figures/images for these sections.
2. Swarm Control 1: Provide all description for the Swarm Control section, including all equation and any theoretical performance guarantee discussion.
3. Swarm Control 2: Provide all results for the Swarm Control section, including figures, images, and tables of the results as well as discussion of those results.
4. Large Language Model 1: Provide all description for the LLM section, including all equation and any theoretical performance guarantee discussion.
5. Large Language Model 2: Provide all results for the LLM section, including figures, images, and tables of the results as well as discussion of those results.
6. Robotic Platform: Provide text and figures/images for this section.
7. Experimental Results: Provide text and figures/images for this section. This student will need to work closely with the rest of the class to obtain the needed results.
8. Video: Create the video. This student will need to work closely with the rest of the class to obtain specific video clips. They are responsible for determining what clips are needed and how to combine them into the final result.